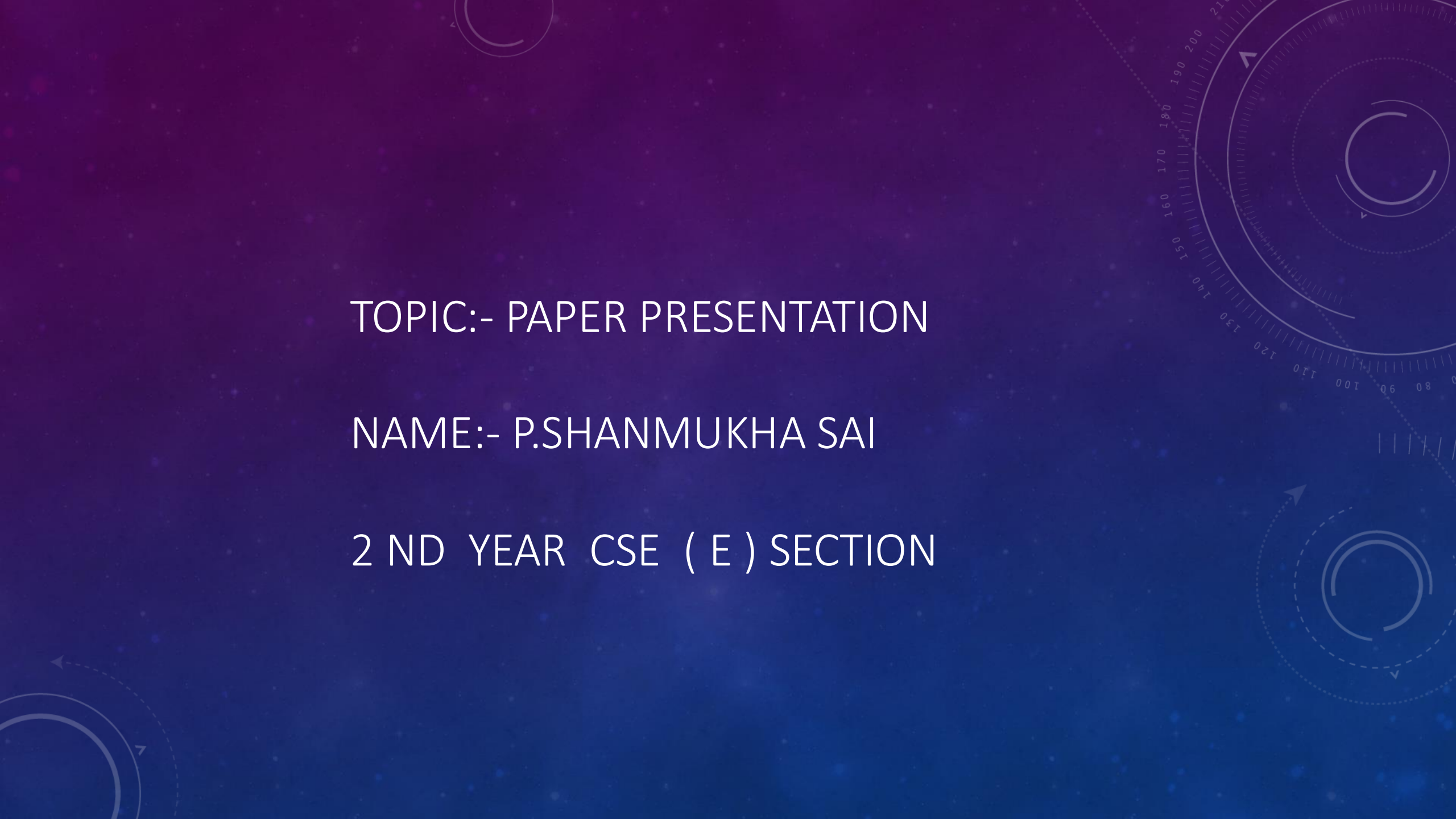




TOPIC:- PAPER PRESENTATION

NAME:- P.SHANMUKHA SAI

2 ND YEAR CSE (E) SECTION



CYBER CRIME AND ETHICAL HACK

Why Cybersecurity MATTERS



Cybersecurity isn't just about corporations or governments, **it's about you and your digital footprint**. Every individual today is a **potential target** because personal data has real monetary and identity value. **Here's why it matters:**

1 Identity Theft



Cybercriminals **steal personal information** (name, NIK, ID, bank account, etc.) to **commit fraud** or **open credit accounts** in your name.

2 Financial Losses

Phishing scams, **fake investment ads**, and **fraudulent apps** can **drain bank accounts** or **steal card data** in seconds.



3 Privacy Invasion



Your **browsing history**, **photos**, or **conversations** can be **tracked**, **sold**, or **leaked** **without consent**.

4 Home Security Risks

Smart devices (CCTV, Wi-Fi routers, IoT appliances) can be **hijacked to spy**, **steal data**, or **launch other attacks**.



5 Chain Reaction

One **compromised individual** often **leads to broader breaches**. **Friends**, **workplaces**, and **online communities** become the next targets.



KEY ASPECTS OF CYBER SECURITY & ETHICAL HACKING

PURPOSE: PROTECT SENSITIVE DATA, PREVENT UNAUTHORIZED ACCESS, AND STRENGTHEN SECURITY INFRASTRUCTURE.

KEY ROLES: ETHICAL HACKERS (WHITE HAT), MALICIOUS HACKERS (BLACK HAT), AND HYBRID HACKERS (GREY HAT).

METHODOLOGIES: VULNERABILITY ASSESSMENT, PENETRATION TESTING, SOCIAL ENGINEERING, AND NETWORK SECURITY.

PHASES OF HACKING: RECONNAISSANCE, SCANNING, GAINING ACCESS, MAINTAINING ACCESS, AND CLEARING TRACKS.

CAREER: HIGH-DEMAND, WELL-COMPENSATED POSITIONS, SUCH AS CERTIFIED ETHICAL HACKERS (CEH).

DARKSIDE



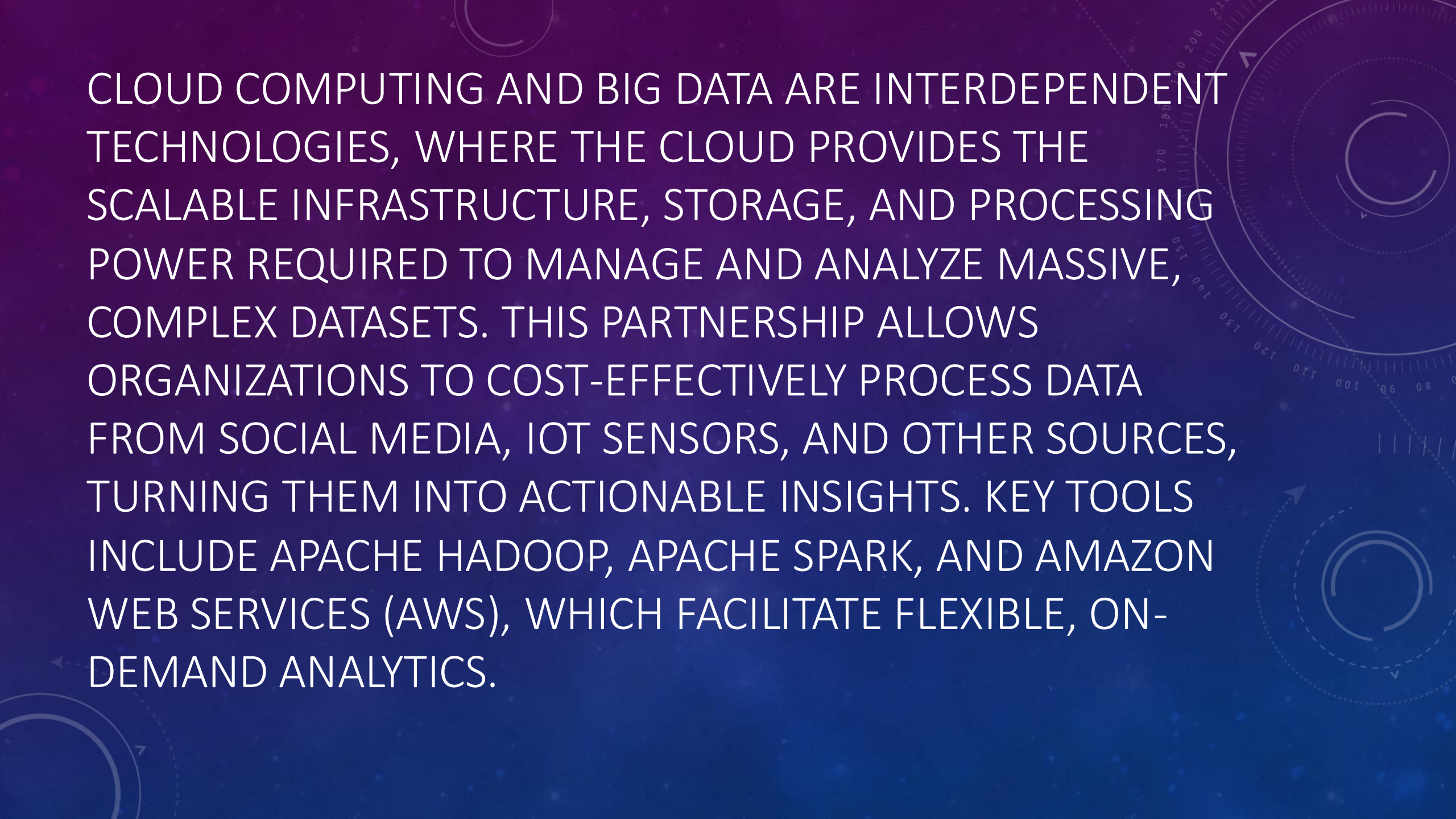
Next
Topic

The image features the words "Next" and "Topic" in a bold, orange, cursive-style font. The word "Next" is positioned above "Topic". A large, orange arrow points from the right side of "Next" towards the right side of "Topic", passing behind the letter "T". The entire graphic is set against a solid orange background. White paint splatters are scattered around the letters, giving it a hand-painted or spray-painted appearance.

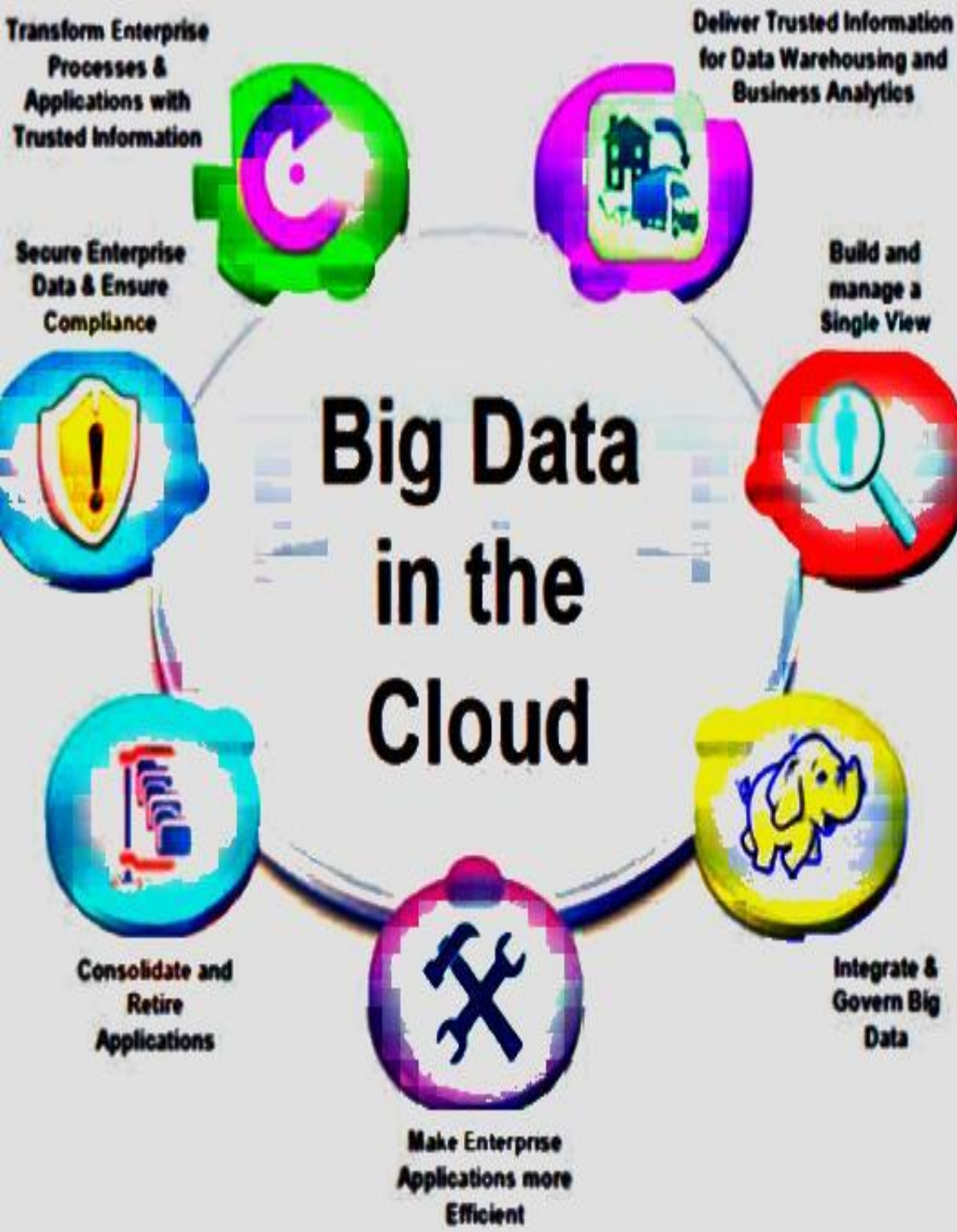


Big Data

**Cloud
Computing**



CLOUD COMPUTING AND BIG DATA ARE INTERDEPENDENT TECHNOLOGIES, WHERE THE CLOUD PROVIDES THE SCALABLE INFRASTRUCTURE, STORAGE, AND PROCESSING POWER REQUIRED TO MANAGE AND ANALYZE MASSIVE, COMPLEX DATASETS. THIS PARTNERSHIP ALLOWS ORGANIZATIONS TO COST-EFFECTIVELY PROCESS DATA FROM SOCIAL MEDIA, IOT SENSORS, AND OTHER SOURCES, TURNING THEM INTO ACTIONABLE INSIGHTS. KEY TOOLS INCLUDE APACHE HADOOP, APACHE SPARK, AND AMAZON WEB SERVICES (AWS), WHICH FACILITATE FLEXIBLE, ON-DEMAND ANALYTICS.



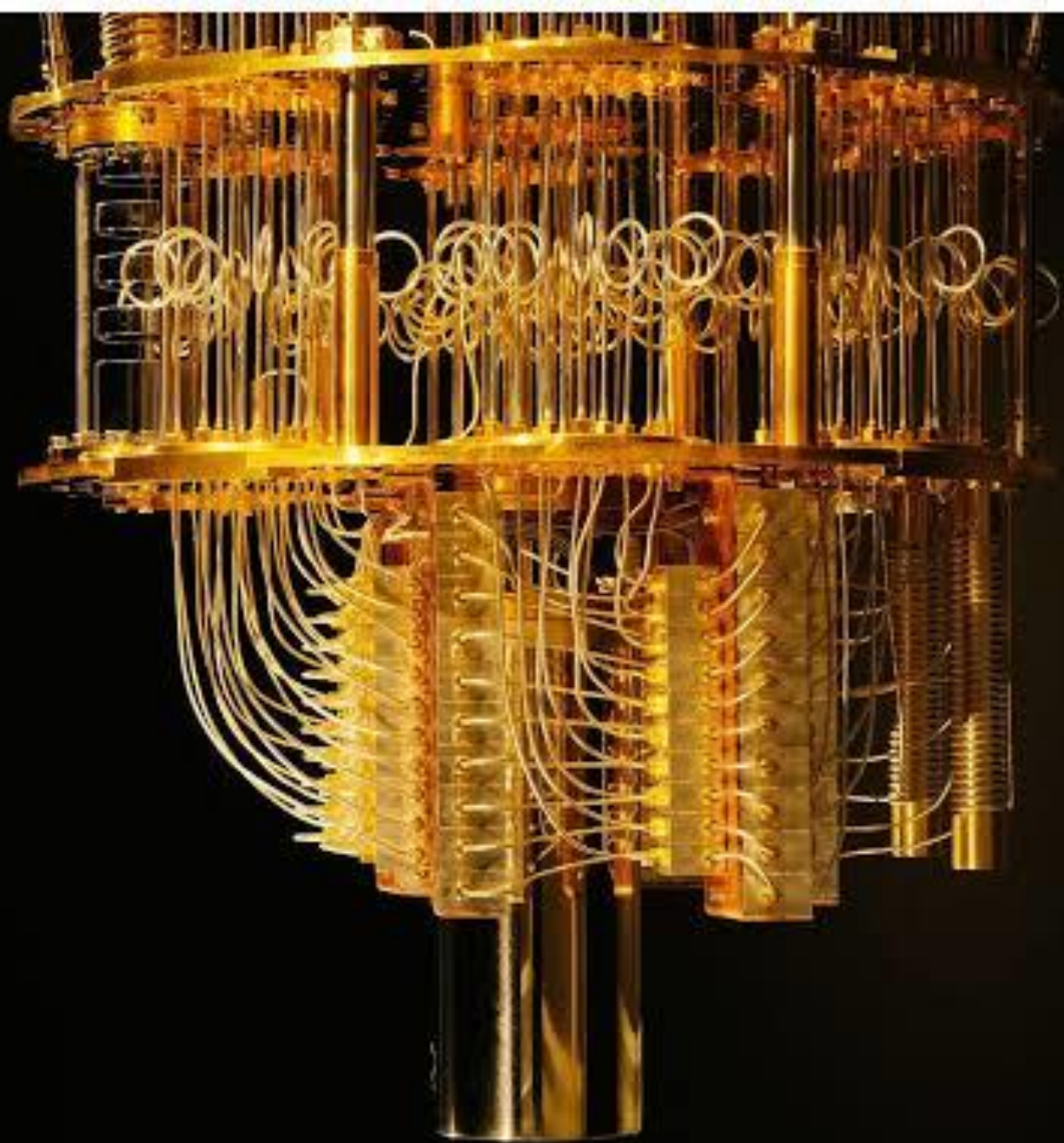
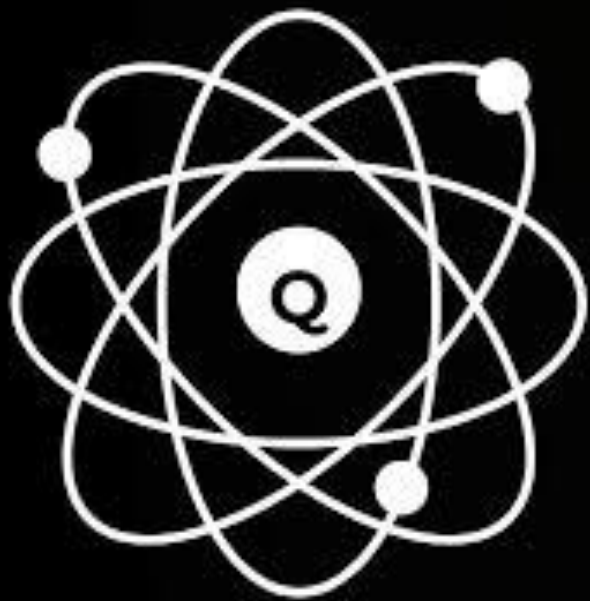
Key Aspects of Cloud Computing and Big Data

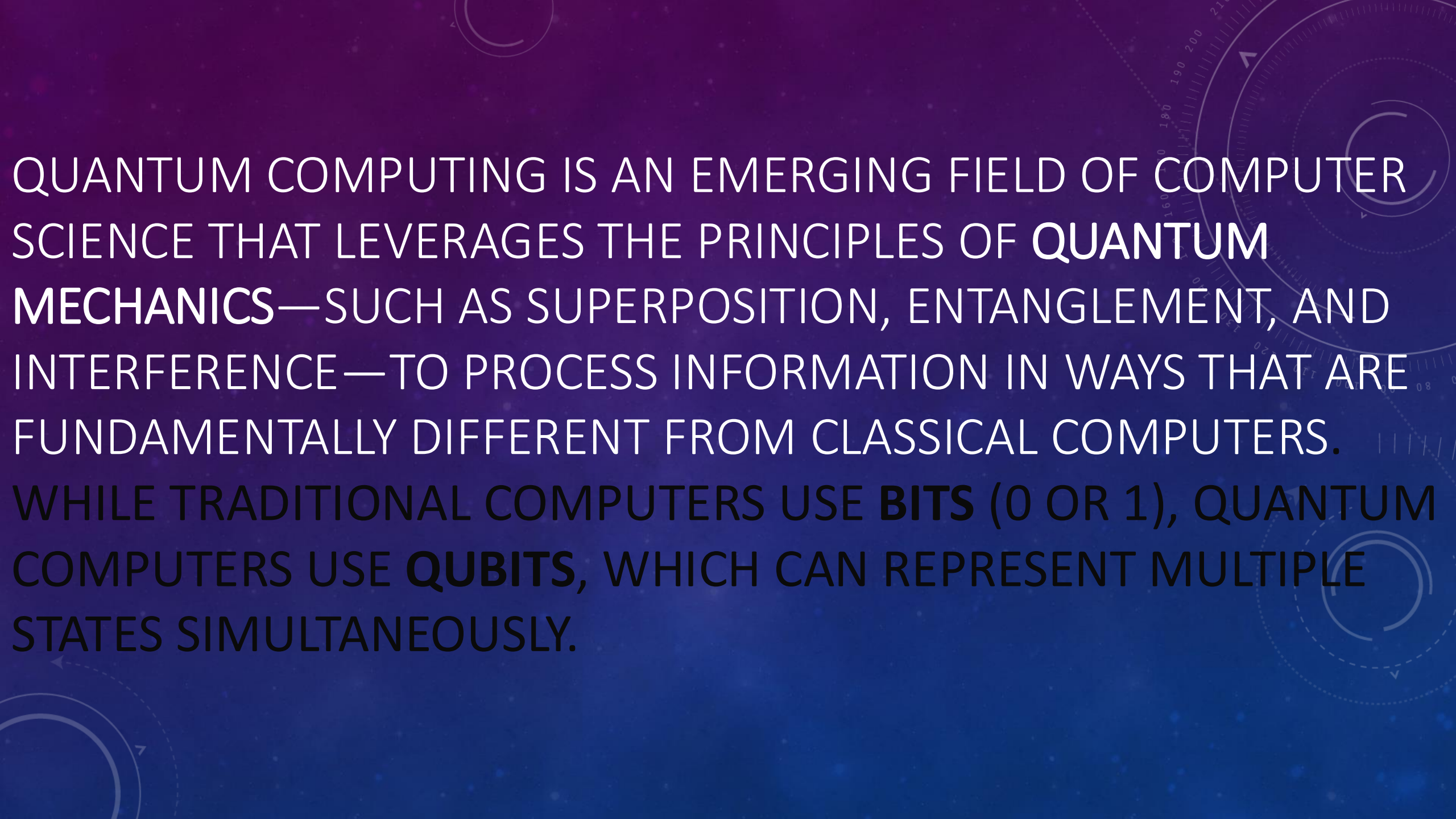
- **Complementary Nature:** Big Data brings the data, while Cloud Computing provides the computational power to handle it.
- **Benefits:** This combination offers unparalleled scalability, reduced infrastructure costs, improved data integration from multiple sources, and enhanced security.
- **Key Drivers:** Technologies such as IaaS, PaaS, and SaaS enable organizations to store and analyze information without managing physical, on-premises servers.
- **Applications:** Common use cases include high-performance data analytics, machine learning, AI model training, and scalable app hosting. 

Next
Topic

The image features the words "Next" and "Topic" in a vibrant blue, bubbly, cursive-style font. The text is outlined in black and has several black paint splatters around it, giving it a dynamic, hand-painted appearance. A large, solid blue arrow points from the word "Next" towards the word "Topic", which is positioned below it. The entire graphic is set against a solid blue background.

Quantum Computing





QUANTUM COMPUTING IS AN EMERGING FIELD OF COMPUTER SCIENCE THAT LEVERAGES THE PRINCIPLES OF **QUANTUM MECHANICS**—SUCH AS SUPERPOSITION, ENTANGLEMENT, AND INTERFERENCE—TO PROCESS INFORMATION IN WAYS THAT ARE FUNDAMENTALLY DIFFERENT FROM CLASSICAL COMPUTERS. WHILE TRADITIONAL COMPUTERS USE **BITS** (0 OR 1), QUANTUM COMPUTERS USE **QUBITS**, WHICH CAN REPRESENT MULTIPLE STATES SIMULTANEOUSLY.



•**Drug Discovery and Healthcare:** Accelerating the simulation of molecular interactions to develop new drugs, such as targeting cancer cells, and enhancing genetic research.

•**Material Science and Chemistry:** Modeling quantum systems to create new, more efficient batteries, stronger materials, or better catalysts for industrial processes.

•**Optimization and Logistics:** Solving complex logistical issues, such as supply chain management, traffic flow optimization, and scheduling.

•**Financial Modeling:** Improving market trend predictions, portfolio optimization, and risk analysis.

•**Cybersecurity and Cryptography:** Developing advanced, hack-resistant encryption algorithms (like Quantum Key Distribution - QKD).

•**Artificial Intelligence (AI):** Enhancing machine learning algorithms for better data analysis and pattern recognition.

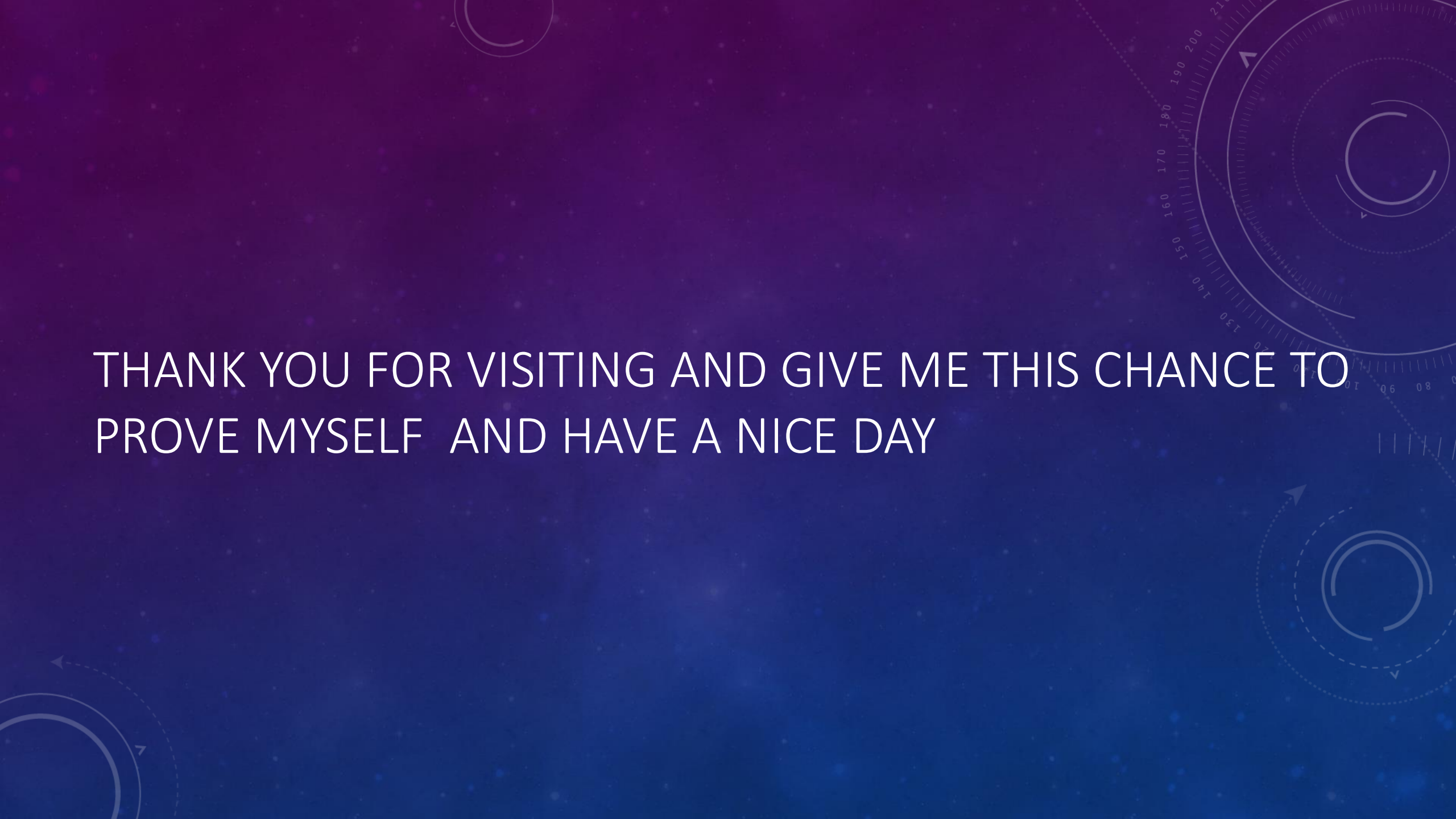
•**Environmental Modeling:** Simulating complex climate systems to create better climate change models and carbon capture materials.



DISADVANTAGES OF QUANTUM COMPUTING

- **Core Disadvantages**

- **Extreme Fragility (Decoherence):** Quantum bits (qubits) are incredibly sensitive to their environment. Even minor vibrations, temperature changes, or electromagnetic waves can cause **decoherence**, where the qubit loses its quantum state and the data is destroyed.
- **High Error Rates:** Unlike classical bits, qubits are prone to high levels of noise and operational faults. Current Noisy Intermediate-Scale Quantum (NISQ) machines require massive redundancy—often thousands of physical qubits to create a single reliable "logical" qubit—to correct these errors.
- **Infrastructure Requirements:** Most quantum processors only function at temperatures near **absolute zero** (-273.15°C), which is colder than outer space. Maintaining this requires massive, expensive cryogenic cooling systems.
- **Prohibitive Costs:** Building and operating a quantum computer can cost hundreds of millions of dollars. This includes specialized hardware like dilution refrigerators and rare materials like **Helium-3** for cooling.
- **Cybersecurity Threats:** Powerful quantum computers could theoretically break modern encryption (like RSA and ECC) that secures the internet. This has led to the risk of "**Harvest Now, Decrypt Later**" attacks, where hackers steal encrypted data today to unlock it once quantum technology matures.
- **Limited Software and Talent:** There is a severe global shortage of experts who can develop quantum algorithms. Furthermore, quantum software is often unique to specific hardware, making it difficult to port or standardize across different systems.



THANK YOU FOR VISITING AND GIVE ME THIS CHANCE TO
PROVE MYSELF AND HAVE A NICE DAY